Supplementary material

How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

Section, table, figure and equation numbers without the prefix S refer to the main text. Every table is generated from the stored results by the scripts named in the text, which are available with the code.

S1. Datasets and preprocessing

All datasets are public. Each recording is cut into non-overlapping windows of T samples that never cross a label change, a time gap or a missing value; windows that would are dropped. Every channel is a separate series. Table S1 gives the resulting sample sizes.

- **WISDM** [1]. Smartwatch accelerometer (x, y, z, m/s^2) at 20 Hz, 51 subjects and five activities (walking, jogging, stairs, sitting, standing). One recording per maximal run of constant subject and activity; timestamps are not used. Group: subject.
- **PAMAP2** [2]. The ± 16 g accelerometers of the three inertial units (hand, chest, ankle; nine channels) at 100 Hz, nine subjects and the twelve protocol activities. Transient periods (activity 0), the heart rate and the saturating ± 6 g accelerometers are not used. Gaps of up to ten samples are filled by linear interpolation; windows with longer gaps are dropped. Group: subject.
- **ESC-50** [3]. 2 000 five-second clips in 50 classes, mono 16-bit PCM scaled to $[-1, 1]$ and resampled from 44.1 to 8 kHz with an anti-aliased polyphase filter. Group: the Freesound source recording, so clips cut from one recording never fall on both sides of a split. The audio study uses the official ESC-10 subset (400 clips) at 2, 4 and 8 kHz and removes windows with a standard deviation below 10^{-3}, which are silent padding.
- **PTB-XL** [4]. Ten-second 12-lead electrocardiograms in mV, read from the 100 Hz files with each lead's gain and baseline; 3 000 of the 21 799 records are drawn at random (seed 0). Labels are the five diagnostic superclasses, as a multilabel task. Group: patient. The sampling-rate study reads the same records from the 500 Hz files.

- **Swarm** [5]. Vector magnetic field B_{NEC} (nT) of the MAGx_LR_1B product at 1 Hz, satellites A, B and C, 1–31 January 2023. Samples whose quality flag is 255 or has bit 2 or 16 set are treated as missing. Unlabelled; group: day.
- **STEAD** [6]. Three-component seismograms (E, N, Z) of 60 s at 100 Hz from the first two data chunks; a class-balanced random subset of 1 000 traces per class (local earthquake, noise) is drawn with seed 0. Group: the earthquake event, or the station for noise, so neither an event nor a noise station falls on both sides of a split.
- **EMG** [7]. Eight-channel armband surface EMG, 36 subjects, static gestures of 3 s. Pauses (label 0) and the extended-palm gesture (label 7, performed by few subjects) are excluded, leaving six gestures. The logged time advances in 1 ms steps by repeating values while the signal changes about 100 times per second, so each file is resampled by sample-and-hold onto a uniform 100 Hz grid. Group: subject.

Table S1: Sample sizes. Fidelity: windows drawn per dataset and length (seed 0) and the non-constant channel series they contain. Classification: windows drawn per dataset and length; ESC-50 is evaluated as ESC-10 instead (Table S11) and Swarm is unlabelled. "–": fewer than 200 windows available.

	Fidelity: windows / channel series			Classification: windows		
Dataset	$T = 64$	$T = 128$	$T = 256$	$T = 64$	$T = 128$	$T = 256$
WISDM	500 / 1500	500 / 1500	500 / 1500	6000	6000	4073
PAMAP2	500 / 4500	500 / 4500	500 / 4500	6000	6000	6000
ESC-50	3000 / 2704	3000 / 2704	3000 / 2699	–	–	–
PTB-XL	500 / 6000	500 / 6000	500 / 6000	6000	6000	6000
Swarm	500 / 1500	500 / 1500	500 / 1500	–	–	–
STEAD	500 / 1494	500 / 1488	500 / 1490	6000	6000	6000
EMG	500 / 4000	500 / 4000	–	1964	880	–

S2. Classifier settings

All classifiers are trained on original windows, standardised per channel with the statistics of the training fold, and evaluated on the original and on every reconstruction of the same held-out windows. KNN ($k = 5$), an RBF

SVM ($C = 1$, Platt probabilities, one-vs-rest for the multilabel PTB-XL) and a random forest (300 trees) take the flattened window. The LSTM and GRU (64 and 32 units, dropout 0.2) and a one-dimensional CNN (convolutions of 32 and 64 filters of width 5, max pooling, global average pooling and a 64-unit dense layer) take the (T, C) window and are trained with Adam, batch size 64 and at most 20 epochs, with early stopping (patience 4) on the last 10% of the training fold. No hyperparameter is tuned. Metrics are computed per fold and averaged over the three folds; the equivalence intervals resample the groups of the split.

S3. Fixed method parameters

No parameter of the encoders or decoders is tuned on the evaluated windows except where stated. The GADF interval $[-0.7, 0.7]$ follows from the branch condition $\alpha < 1/\sqrt{2}$ (Section 3). The number of GADF pivots (five) and of MTF states ($Q = 4$) were fixed in our earlier single-dataset study on WISDM, which is also one of the seven datasets here, and were not changed for this benchmark. Hirata's recurrence rate (0.75) was fixed before the benchmark; its sensitivity is reported in Section S6. Sipers' frequency threshold (10^{-4}) was first chosen in an exploratory sweep on evaluated windows (Section S7) and confirmed on calibration windows outside the evaluated samples (Section S8).

S4. Fidelity at the other window lengths

Tables S2 and S3 repeat Table 3 of the main text for $T = 64$ and $T = 256$. EMG at $T = 256$ has fewer than 200 windows and is not reported.

Table S2: Mean 8-bit NMAE of every encoding–decoder combination at $T = 64$. Uns./Sig.: unsigned and signed RP.

	Hirata		Sipers		Geometric (ours)					
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2647	0.1171	0.2084	0.0042	0.2039	0.0012	0.0016	0.0011	0.2930	0.0003
PAMAP2	0.2630	0.1153	0.2108	0.0049	0.2103	0.0017	0.0017	0.0012	0.2889	0.0003
ESC-50	0.2573	0.0959	0.2104	0.0056	0.2087	0.0018	0.0018	0.0010	0.2777	0.0003
PTB-XL	0.3530	0.1916	0.2991	0.0060	0.2910	0.0022	0.0033	0.0019	0.3434	0.0003
Swarm	0.2710	0.0293	0.2613	0.0028	0.2630	0.0144	0.0025	0.0013	0.2942	0.0003
STEAD	0.2361	0.1001	0.2052	0.0055	0.2030	0.0014	0.0016	0.0010	0.2733	0.0003
EMG	0.2300	0.1365	0.1869	0.0039	0.1947	0.0010	0.0014	0.0020	0.2982	0.0006

Table S3: As Table S2, at $T = 256$.

	Hirata		Sipers		Geometric (ours)					
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2228	0.0971	0.1743	0.0046	0.1682	0.0010	0.0014	0.0015	0.2700	0.0002
PAMAP2	0.2084	0.0960	0.1829	0.0053	0.1736	0.0015	0.0015	0.0015	0.2740	0.0002
ESC-50	0.1987	0.0755	0.1747	0.0058	0.1759	0.0017	0.0015	0.0011	0.2672	0.0001
PTB-XL	0.3066	0.1600	0.2588	0.0051	0.2716	0.0012	0.0025	0.0026	0.3291	0.0002
Swarm	0.2950	0.0339	0.3111	0.0155	0.2573	0.0174	0.0027	0.0009	0.3066	0.0001
STEAD	0.1723	0.0789	0.1581	0.0047	0.1552	0.0012	0.0014	0.0011	0.2592	0.0001
EMG	–	–	–	–	–	–	–	–	–	–

S5. Spectral and correlational fidelity

Table S4 separates the two reasons for a large pointwise error discussed in Section 6.1: decoders that return the reflection keep the spectrum and the energy, while those that lose the waveform distort them.

Table S4: Spectral and correlational fidelity after 8-bit storage: range of the dataset means over the seven datasets at $T = 128$. Pearson: correlation with the original; PSD shape: L_1 distance between unit-mass Welch spectra; Energy: relative error of total power.

Decoder	Pearson	PSD shape	Energy
Hirata / uns.	-0.023–0.019	0.0550–0.6117	0.224–0.766
Hirata / sig.	0.556–0.982	0.0562–0.6053	0.225–0.730
Sipers / uns.	-0.075–0.038	0.0021–0.0125	0.005–0.012
Sipers / sig.	0.999–1.000	0.0063–0.0260	0.010–0.020
Ours / uns.	-0.036–0.045	0.0007–0.0151	0.001–0.036
Ours / sig.	0.999–1.000	0.0034–0.0187	0.002–0.055
GASF	0.999–1.000	0.0041–0.0171	0.005–0.012
GADF	1.000–1.000	0.0005–0.0048	0.001–0.002
MTF	-0.021–0.009	0.3279–0.7871	1.053–3.154
XIRP	1.000–1.000	0.0003–0.0017	0.000–0.001

S6. Recurrence-rate sensitivity of Hirata's reconstruction

Table S5 gives the 8-bit NMAE of Hirata's reconstruction [8] against the recurrence rate used to threshold the image ($T = 128$, 100 benchmark windows per dataset, every channel). The rate of 0.75 used in the main text was fixed in advance; choosing the best rate per dataset with knowledge of the result lowers the error by at most 0.025 on the unsigned and 0.010 on the

signed RP. Because this sweep uses evaluated windows, it is reported as a sensitivity analysis, not used to choose the rate.

Table S5: Hirata's reconstruction: 8-bit NMAE against the recurrence rate.

Dataset	Encoding	0.02	0.05	0.1	0.2	0.3	0.5	0.7	0.75	Best
WISDM	unsigned	0.239	0.250	0.248	0.245	0.239	0.240	0.237	0.231	0.75
	signed	0.225	0.196	0.171	0.136	0.120	0.103	0.104	0.103	0.75
PAMAP2	unsigned	0.240	0.255	0.252	0.238	0.236	0.231	0.233	0.238	0.5
	signed	0.222	0.194	0.161	0.124	0.105	0.098	0.102	0.107	0.5
ESC-50	unsigned	0.243	0.251	0.235	0.222	0.226	0.209	0.202	0.197	0.75
	signed	0.223	0.156	0.118	0.084	0.080	0.075	0.081	0.083	0.5
PTB-XL	unsigned	0.307	0.311	0.320	0.323	0.326	0.330	0.342	0.328	0.02
	signed	0.290	0.287	0.285	0.277	0.272	0.252	0.213	0.193	0.75
Swarm	unsigned	0.273	0.265	0.287	0.291	0.293	0.296	0.293	0.290	0.05
	signed	0.064	0.033	0.021	0.020	0.024	0.028	0.028	0.028	0.2
STEAD	unsigned	0.239	0.251	0.235	0.222	0.224	0.219	0.221	0.212	0.75
	signed	0.216	0.162	0.124	0.096	0.085	0.079	0.084	0.085	0.5
EMG	unsigned	0.214	0.234	0.234	0.220	0.215	0.201	0.200	0.203	0.7
	signed	0.199	0.192	0.163	0.138	0.126	0.116	0.119	0.121	0.5

S7. Sipers et al. and the geometric decoder on the same image

The reconstruction of Sipers et al. [9] assumes a periodic signal and a circular delay embedding, so in the main benchmark it decodes the circular variant of the RP, while the geometric decoder decodes the conventional RP. The circular RP is built on the states (x_i, x_{i+1}), $i = 1, \ldots, T$, with $x_{T+1} = x_1$: it is the conventional RP with one added state, the wrap state (x_T, x_1), stored with the same 8-bit maps. Sipers' method represents the window by its periodic extension, which a finite sequence always admits, so its hypotheses hold for the circular embedding of any window; the cost is the wrap state, which joins the last sample to the first and is not part of the recorded dynamics.

Table S6 removes the difference of encoding: both decoders receive the identical circular image of the first 100 benchmark windows of each dataset at $T = 128$, every channel, with the same per-channel calibration, and on the signed RP both choose the reflection from the image alone with the same criterion, so any gap is due to the decoder. The unsigned columns fix the reflection with the original series; they are a bound on what the geometry alone allows, not a reconstruction.

Sipers' decoder discards frequencies whose power, estimated from the image, is below a share of the largest; this frequency threshold does not use

the original series. With a threshold of zero the implementation is exact in floating point (median error 0; maximum 2e-13 on WISDM, Swarm and STEAD, and 2.7e-03 for the worst of 1 200 PTB-XL series, where near-zero Fourier coefficients enter the division of Eq. (15) of Sipers et al.). With the threshold of 0.01 used by its authors, the error is 0.009 to 0.044 even in floating point, because our windows are broadband. The last columns of Table S6 sweep the threshold on these evaluated windows. This exploratory sweep is where the value used throughout, 10^{-4}, was first chosen, so it is not independent of the evaluation; Section S8 confirms it on separate calibration windows.

Table S6: Mean 8-bit NMAE on the same circular RP ($T = 128$, first 100 benchmark windows per dataset). Unsigned: reflection fixed with the original series, a bound on the geometry alone. Signed: reflection chosen from the image by the sign, as in the main text. Last columns: Sipers' decoder on the unsigned image against the frequency threshold.

	Unsigned, oracle sign		Signed		Sipers, unsigned, oracle sign, frequency threshold				
Dataset	Sipers	Geometric	Sipers	Geometric	0.01	0.001	0.0001	1e-06	0
WISDM	0.0026	0.0004	0.0042	0.0010	0.0440	0.0092	0.0026	0.0031	0.0034
PAMAP2	0.0033	0.0006	0.0050	0.0014	0.0296	0.0082	0.0033	0.0042	0.0043
ESC-50	0.0039	0.0006	0.0059	0.0012	0.0339	0.0107	0.0039	0.0041	0.0041
PTB-XL	0.0035	0.0007	0.0054	0.0014	0.0216	0.0052	0.0035	0.0050	0.0054
Swarm	0.0019	0.0010	0.0046	0.0025	0.0410	0.0379	0.0019	0.0020	0.0020
STEAD	0.0040	0.0005	0.0054	0.0012	0.0271	0.0078	0.0040	0.0055	0.0061
EMG	0.0015	0.0003	0.0039	0.0008	0.0093	0.0016	0.0015	0.0017	0.0017

S8. Calibration of Sipers' frequency threshold

The threshold first chosen in Section S7 is checked on 100 windows per dataset at $T = 128$ that lie outside both the fidelity and the classification samples, drawn with a different seed (seed 1; script `_sipers_calibration.py`). Such windows exist for 6 of the seven datasets; all EMG windows are evaluated. Averaging the 8-bit NMAE over datasets with equal weight, the threshold with the lowest error is 0.0001 when the reflection is fixed with the original series and 0.0001 when it is chosen from the signed image, the value used throughout. The calibration windows may come from the same recordings as evaluated windows.

S9. Sampling rate at a fixed length and at a fixed duration

Table S7 separates the sampling rate from the window duration on PTB-XL. With 128 samples, a higher rate also means a shorter window; with a fixed duration of 1.28 s, the number of samples grows with the rate. In both settings the conditioning σ_2/σ_1 falls and the error of the signed RP rises with the rate, while GASF, GADF and XIRP do not follow that trend. The "native" rows use the 100 Hz files distributed with the dataset instead of resampling the 500 Hz records.

Table S7: PTB-XL, anti-aliased resampling from 500 Hz: median σ_2/σ_1 of the RP states, share of series below 0.1, and mean 8-bit NMAE.

Window	Rate	Seconds	σ_2/σ_1	Share < 0.1	Signed RP	Sipers / sig.	GASF	GADF	XIRP
128 samples	500 Hz	0.26	0.08	67%	0.0076	0.0078	0.0032	0.0018	0.0005
	250 Hz	0.51	0.16	23%	0.0041	0.0074	0.0032	0.0019	0.0002
	100 Hz	1.28	0.37	0%	0.0015	0.0055	0.0029	0.0022	0.0002
	100 (native) Hz	1.28	0.37	0%	0.0015	0.0054	0.0028	0.0022	0.0002
	50 Hz	2.56	0.61	0%	0.0009	0.0040	0.0021	0.0019	0.0002
1.28 s	500 Hz	1.28	0.08	76%	0.0079	0.0061	0.0028	0.0032	0.0002
	250 Hz	1.28	0.16	9%	0.0026	0.0064	0.0028	0.0028	0.0002
	100 Hz	1.28	0.37	0%	0.0015	0.0055	0.0029	0.0022	0.0002
	100 (native) Hz	1.28	0.37	0%	0.0015	0.0054	0.0028	0.0022	0.0002
	50 Hz	1.28	0.64	0%	0.0012	0.0037	0.0024	0.0016	0.0003

S10. Robustness per dataset

Table S8 gives, for each dataset, the clean 8-bit error and the error with Gaussian noise of $\sigma = 16$ intensity levels on the stored image, which Figure 4a averages over datasets.

Table S8: Mean 8-bit NMAE at $T = 128$: clean / with noise of $\sigma = 16$ (200 windows per dataset, two replicates). Sipers' decoder reads the circular RP.

Dataset	Ours / sig.	Sipers / sig.	GASF	GADF	XIRP
WISDM	0.0010 / 0.036	0.0043 / 0.083	0.0015 / 0.062	0.0013 / 0.066	0.0002 / 0.012
PAMAP2	0.0015 / 0.044	0.0049 / 0.082	0.0016 / 0.065	0.0013 / 0.064	0.0002 / 0.011
ESC-50	0.0017 / 0.042	0.0059 / 0.084	0.0016 / 0.062	0.0010 / 0.062	0.0002 / 0.011
PTB-XL	0.0016 / 0.051	0.0054 / 0.093	0.0030 / 0.074	0.0023 / 0.068	0.0002 / 0.011
Swarm	0.0126 / 0.062	0.0044 / 0.084	0.0026 / 0.063	0.0015 / 0.060	0.0002 / 0.012
STEAD	0.0012 / 0.032	0.0050 / 0.080	0.0014 / 0.061	0.0010 / 0.064	0.0002 / 0.011
EMG	0.0009 / 0.022	0.0040 / 0.075	0.0013 / 0.061	0.0022 / 0.063	0.0006 / 0.011

S11. Classification per dataset

Table S9 gives the score of the classifiers on the original windows at $T = 128$, and Table S10 the drop of each combination per dataset, averaged over classifiers and lengths; Table 4 of the main text aggregates them. Table S11 gives the per-clip results on ESC-10.

Table S9: Score on the original windows at $T = 128$: accuracy, or macro-F1 for the multilabel PTB-XL.

Dataset	KNN	SVM	RF	LSTM	GRU	CNN
WISDM	0.467	0.722	0.750	0.652	0.709	0.746
PAMAP2	0.540	0.651	0.689	0.541	0.596	0.589
PTB-XL	0.299	0.328	0.154	0.525	0.593	0.571
STEAD	0.511	0.490	0.658	0.517	0.511	0.583
EMG	0.166	0.409	0.770	0.652	0.675	0.813

Table S10: Drop from the original score, averaged over the six classifiers and the three lengths.

Dataset	Hirata / uns.	Hirata / sig.	Sipers / uns.	Sipers / sig.	Ours / uns.	Ours / sig.	GASF	GADF	MTF	XIRP
WISDM	0.087	0.027	0.066	0.001	0.068	0.000	0.000	0.000	0.069	0.000
PAMAP2	0.146	0.119	0.115	0.000	0.125	0.000	-0.001	0.000	0.115	0.000
PTB-XL	0.221	0.152	0.215	0.000	0.210	0.000	-0.001	0.000	0.223	0.000
STEAD	0.007	0.002	0.004	0.000	0.004	0.000	0.000	0.000	0.007	0.000
EMG	0.150	0.109	0.082	0.001	0.085	0.000	0.000	0.000	0.074	0.000

Table S11: ESC-10: per-clip accuracy of the original, averaged over the six classifiers, and drop of each combination (3 000 windows per block).

Block	Original	Hirata / uns.	Hirata / sig.	Sipers / uns.	Sipers / sig.	Ours / uns.	Ours / sig.	GASF	GADF	MTF	XIRP
2 kHz, $T = 512$	0.312	0.081	0.069	0.028	0.001	0.026	-0.001	0.000	-0.001	0.125	-0.001
4 kHz, $T = 512$	0.339	0.076	0.063	0.023	-0.003	0.025	0.000	-0.002	-0.002	0.118	0.000
8 kHz, $T = 512$	0.311	0.061	0.043	0.017	-0.001	0.019	0.001	0.001	0.000	0.066	0.000
2 kHz, $T = 1024$	0.324	0.065	0.063	0.020	0.000	0.023	0.000	0.002	0.002	0.145	0.000
4 kHz, $T = 1024$	0.306	0.077	0.072	0.017	0.003	0.026	0.003	0.000	0.001	0.106	0.000
8 kHz, $T = 1024$	0.305	0.065	0.047	0.020	0.005	0.024	0.004	0.002	0.003	0.080	0.000

S12. Computational cost

Table S12 gives the single-thread cost of every combination against the window length. Times were measured on an otherwise idle Intel Core i7-14650HX (12 cores, 24 threads) under WSL2 (Linux 6.6), with Python 3.10.20,

NumPy 1.26.4 and SciPy 1.14.0, one series at a time and multi-threaded linear algebra disabled; classifiers used scikit-learn 1.5.1 and TensorFlow 2.11.0.

Table S12: Median single-thread time (ms) to encode, store and decode one channel (WISDM, 20 series per length), log–log slope of time against T, and peak traced memory at $T = 1024$.

Decoder	$T = 64$	$T = 256$	$T = 1024$	Exponent	MB at 1024
Hirata / uns.	2.0	58.5	2871.2	2.64	87
Hirata / sig.	2.3	71.8	2880.1	2.59	87
Sipers / uns.	1.1	27.1	1172.7	2.47	112
Sipers / sig.	1.5	29.9	1193.1	2.45	112
Ours / uns.	0.6	13.1	398.2	2.35	72
Ours / sig.	1.3	25.1	552.0	2.22	72
GASF	0.1	1.0	12.0	1.85	32
GADF	0.1	1.0	13.0	1.72	32
MTF	0.5	1.5	12.8	1.22	24
XIRP	0.1	0.6	12.8	2.03	32

S13. Proof of Proposition 1

We use the notation of Section 3.2 and the decoder steps of Section 4.2 of the main text. Let $S \in \mathbb{R}^{(T-1)\times 2}$ have rows $s_i = (x_i, x_{i+1})$.

Geometry. By (a), the centred states have affine rank two, so classical multidimensional scaling applied to the exact distance matrix $|R|$ returns $Y = SP_0 + \mathbf{1}t^\top$ for some $P_0 \in O(2)$ and $t \in \mathbb{R}^2$ [10]. Every candidate configuration produced by step 3 is therefore of the form $Z = SP + \mathbf{1}c^\top$ with $P \in O(2)$.

Overlap. Write (a_i, b_i) for the rows of Z. Then

$$b_i - a_{i+1} = P_{12}x_i + (P_{22} - P_{11})x_{i+1} - P_{21}x_{i+2} + \text{const},$$

so the residual is constant in i if and only if $P_{12}x_i + (P_{22} - P_{11})x_{i+1} - P_{21}x_{i+2}$ is. For a rotation, $P = \left(\begin{smallmatrix} c & s \\ -s & c \end{smallmatrix}\right)$, this reads $s(x_i + x_{i+2}) = \text{const}$; for a reflection, $P = \left(\begin{smallmatrix} c & s \\ s & -c \end{smallmatrix}\right)$, it reads $s(x_i - x_{i+2}) - 2c\,x_{i+1} = \text{const}$. Under (b), the first equation forces $s = 0$, and the second has no solution with (s, c) on the unit circle, since it would be relation (F) with $\alpha = s$ and $\beta = -2c$. Hence the only configurations that satisfy the overlap exactly are $P = \pm I$, i.e. the states of x and of $-x$, up to a translation. In step 3 this means that one of the two

matrices H_η has a zero eigenvalue, whose eigenvector, taken with both signs, yields these two candidates; the eigenvalue is simple because a vanishing H_η would make every transformation of that parity a solution. The other matrix has no zero eigenvalue and is positive definite. Conversely, (R) makes every rotation a solution, so one H_η vanishes, and (F) gives a reflection solution, a zero eigenvalue of the other parity, which proves the equivalence stated in the proposition. If the centred triples (x_i, x_{i+1}, x_{i+2}) have rank three, no nontrivial linear relation among three consecutive samples holds for all i, so neither (R) nor (F) can hold. Step 5 removes the translation, and the final range map removes the remaining translation and positive scale.

Sign. In step 4 the candidate x reproduces the observed matrix exactly, after both are normalised to unit Frobenius norm. Any candidate whose signed RP also reproduces it has states with distance matrix proportional to $|R|$ that satisfy the overlap by construction, and is therefore $\pm x$ up to scale and translation by the previous paragraph. Reflecting the series maps $s_i - s_j$ to $s_j - s_i$, so σ_{ij} becomes σ_{ji}: on a sign-informative pair the entry changes sign, and on a non-informative pair it does not. The signed RPs of x and $-x$ therefore differ if and only if (c) holds; under (c) the selected candidate is x, and without (c) the two are indistinguishable from the image. □

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